YAKEEN

NEET 2.0 2027

Physics | Lecture - 01

Motion in a Straight Line

By- Manish Raj (MR Sir)



Topics To Be Covered

Introduction

NO COPY

Physics

11th Physics

- ✓ kinematics (1-D, 2-D)
- ✓ Laws of motion
- ✓ Work energy Power
- ✓ Rotation
- ✓ Gravitation, Solid
- ✓ Fluid
- ✓ Thermal, ✓ Thermo
- ✓ S.H.M ✓ Wave, K+G ✓

25 chapter

7 Mon

= 3.5 chapter Per Month

12th Physics

- ✓ Electric charge field
- ✓ electric Potⁿ, ✓ capacity
- ✓ current electricity
- ✓ Magnetism, MEC
- ✓ Electromagnetic Induⁿ
- ✓ AIC EMI
- ✓ Ray Optics, ✓ Wave optics
- ✓ Modern
- ✓ semiconductor

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Basic math
vectors
or U.d.M. ka backlog
hai to bhi kinematics
se start kar sktehai

Unit & dimn ka backlog
kinte logo ka hai ??

(a) yes, backlog hai (32%)

(b) NO → 67%

for vectors, basic math & Utm
No need watch long lecture

→ 5hr detailed backlog lecture
& do Opp & Sanyukt assignment

Physics :—

Imagination is more important than Knowledge —

Concept :-

↳ relate with daily life :—

enjoy

Numerical

NO Imagination, NO vision

↳ write the data. solve every step :

theory rough page

P3 like ke neeche

Fmive

1. Distance of Delhi railway station is ??

Ans → wrong question, refrence is not given

→ A place from where we take obserbation

obseber → who take obserbation

(2.) distⁿ of delhi railway station from Patna ??

Ans → 1000 KM

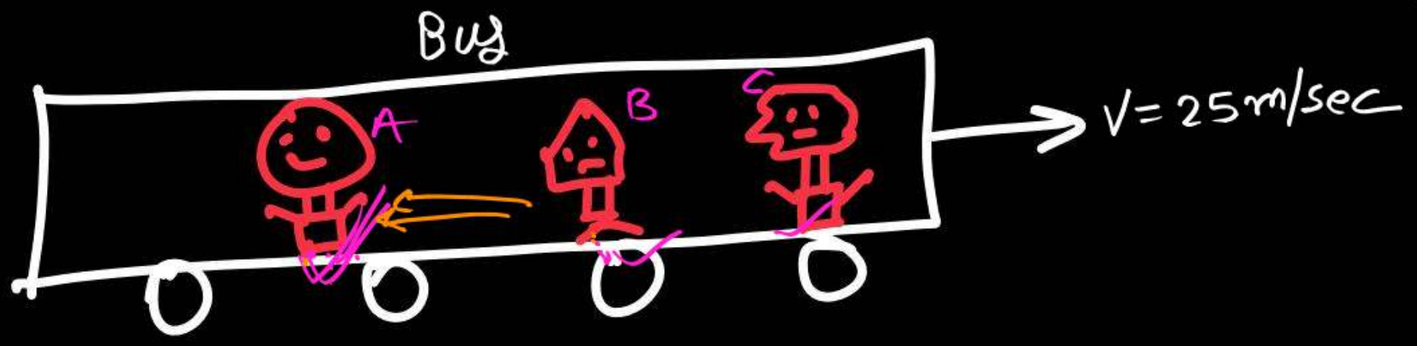
→ depends on Path.



lift → frame of refⁿ



When delhi will come ??





(a) correct (20%) (+1) in terms of physics

(b) wrong (80%) (-1)

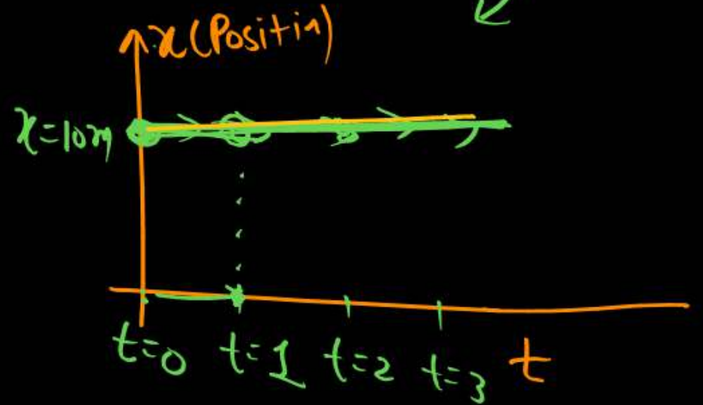
A is at rest w.r.t 'A'



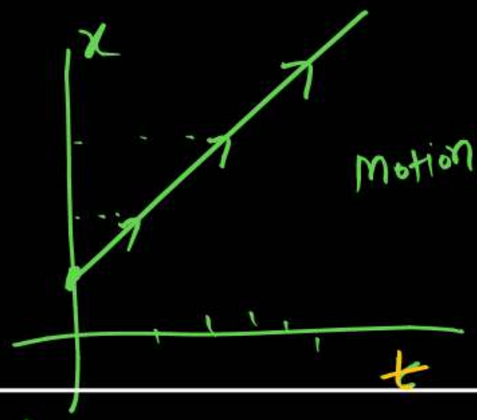
✓ observer always assume to be at rest w.r.t himself" ← copy

Position → Location of object w.r.t frame of refrence.

gf Position is Not Changing w.r.t. Time → Rest



gf Position is changing w.r.t time = Motion ✓



→ Rest & motion is a relative term, relative to frame of refrence.

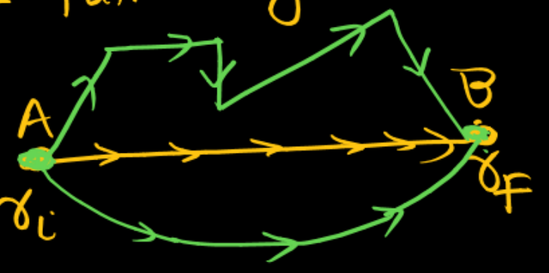
→ Nothing is absolute in this Universe.

Distance

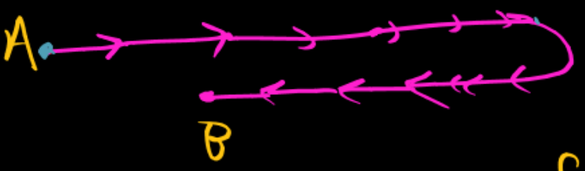
- length of Actual Path covered
- odometre ki reading distance
- always Increasing for moving object.
- unit metre
- Scalar
- distance Can't be negative.
- depends UPon Path taken b/w Two Point.

Displacement

- Shift in Position of moving object.
- $\Delta\vec{r} = \vec{r_f} - \vec{r_i}$
- vector, metre.
- Shotest Path length b/w Two Point



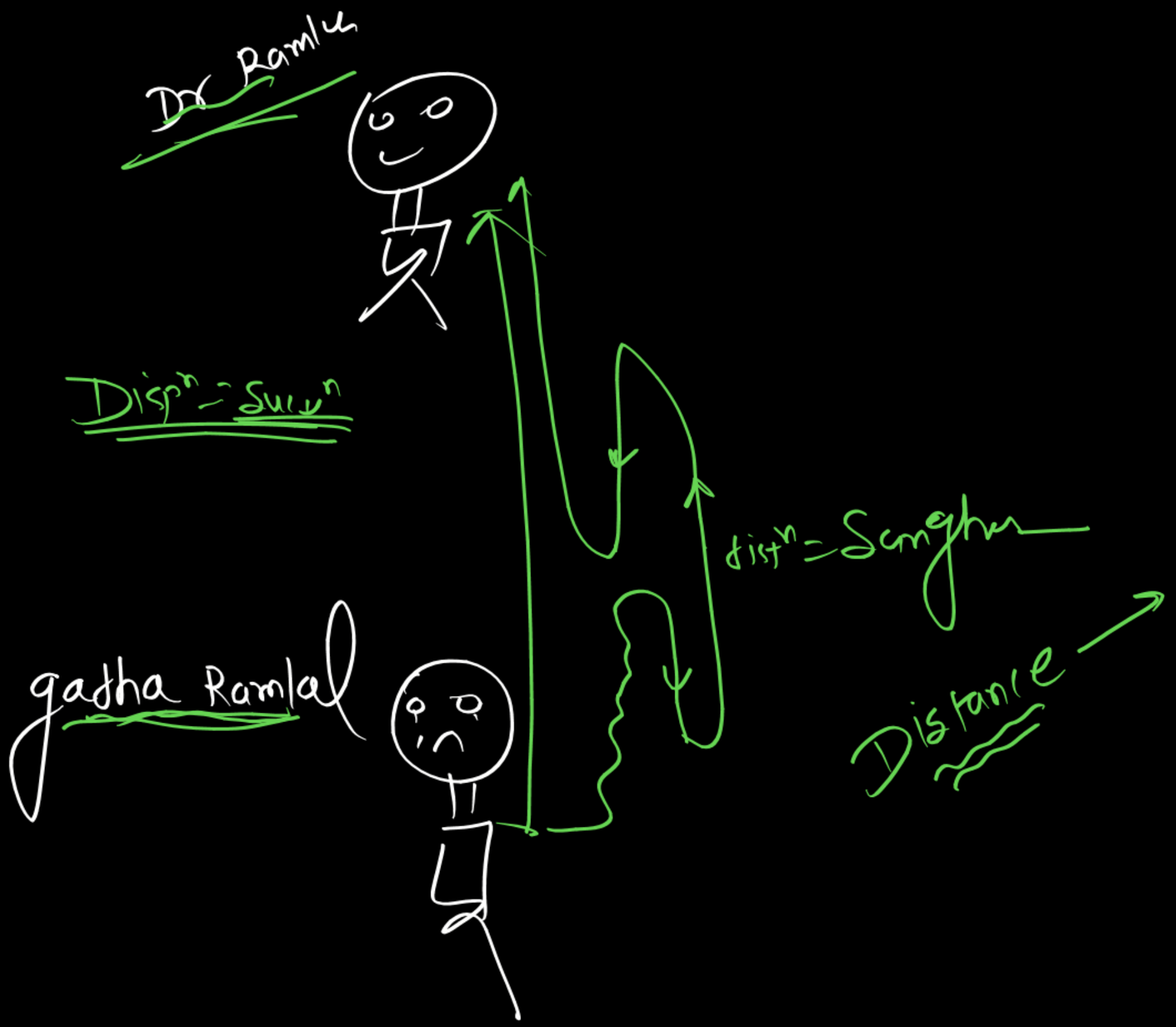
- $disp^m$ does not depends UPon Path taken.



- $disp^m$ can decreasing for moving object
- $disp^m$ Can be +ve & -ve.
- Position Kitna (How much) & Kidhar (dirn) me shift huaa.

MR* Box
- gf intial & final Position same then $disp^m = 0$

MR* Box → gf Path is not Known b/w two Point then you can't calculate distance


Dr Ramlal
Displ^n = Sum^n
gadha Ramlal
dist^n = Sum^n
Distance

If object is moving in one-dimension without change in direction :-

$$|disp^m| = \text{distance}$$

If object is moving in straight line ✓

$$dist^n \geq |disp^m|$$

$$\frac{dist^n}{|disp^m|} \geq 1$$

Copy

aisa koi case jisme $|disp^m| > dist^n$.

↳ Never Possible

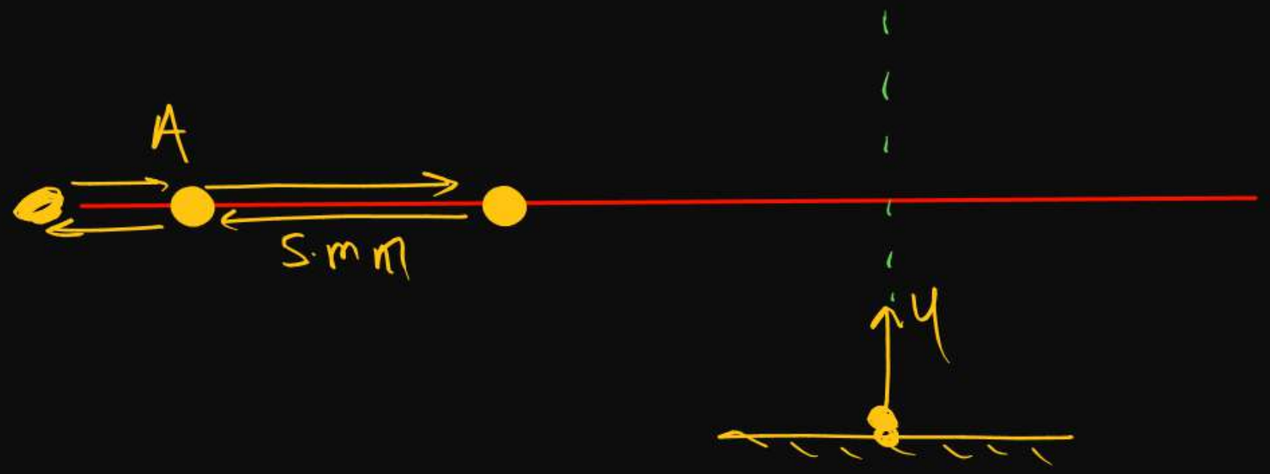
As magnitude of $disp^m$ is distance ⟶ NO, not new

→ Can be say only when direction object is not changing

Question

Which of the following option is correct for motion in Straight line (1-D motion)

1. distance = | disp^m |
2. distance > | disp^m |
3. distance ≥ | disp^m |
4. distance < | disp^m |



Question

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The numerical ratio of distance to the displacement covered is always

1. less than one
2. equal to one
3. equal to or less than one
4. equal to or greater than one

$\frac{\text{distn}}{|\text{disp}^{n}|} \geq 1$

$\therefore \frac{\text{distn}}{|\text{disp}^{n}|} \geq 1$

18%

70%

Question

NO COPY

Which of the following option is correct for motion in 1–D, without change in direction.

1. distance = $|\text{disp}^m|$
2. distance > $|\text{disp}^m|$
3. distance ≥ $|\text{disp}^m|$
4. distance < $|\text{disp}^m|$

Question

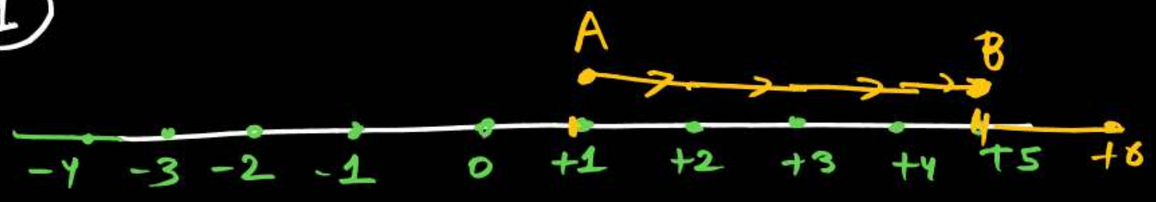
Copy

Which of the following option is correct for motion in 1-D, with [redacted] change in direction.

1. distance = | disp^m |
2. distance > | disp^m | (40.1)
3. distance ≥ | disp^m | (54.1)
4. distance < | disp^m |



(1)



$dist^n = 4m$

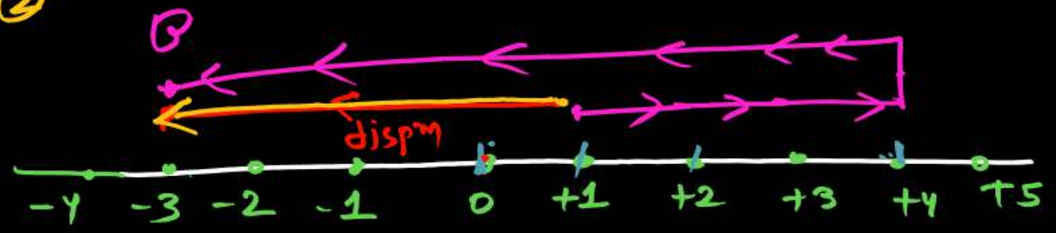
$\underline{disp^m} = 5\hat{i} - 1\hat{i}$

$= +4\hat{i}.$

$\underline{disp^m} = \vec{r}_f - \vec{r}_i$

shift in position

(2)

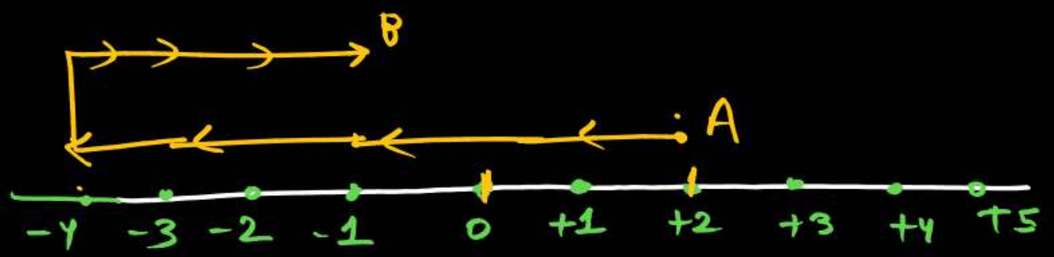


$dist^n = 3+4+3 = 10m$ ✓

$disp^m = -3\hat{i} - (1\hat{i})$

$= -4\hat{i}\,m$

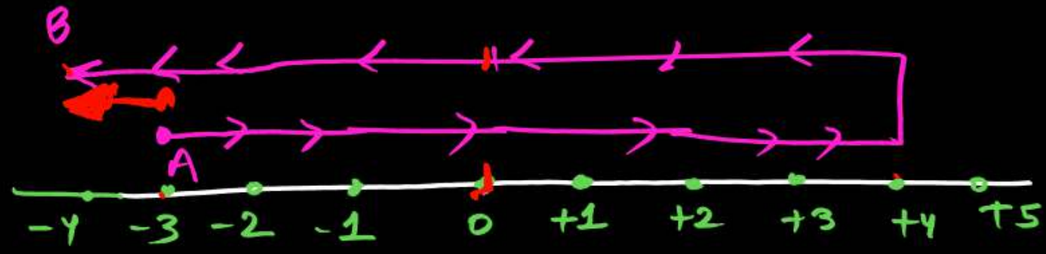
(3)



$dist^n = 2+4+3 = 9m$

$disp^n = -1\hat{i} - 2\hat{i} = -3\hat{i}$

(4)



$dist^m = 3+4+4+4 = 15m$

$disp^m = -4\hat{i} - (-3\hat{i})$

$= -1\hat{i}\,m$

(5) intial Position $(\underline{2,4,7})$ & final Position $(\underline{5,8,12})$ then find disp^n

$\vec{r}_f$ $\vec{r}_i$



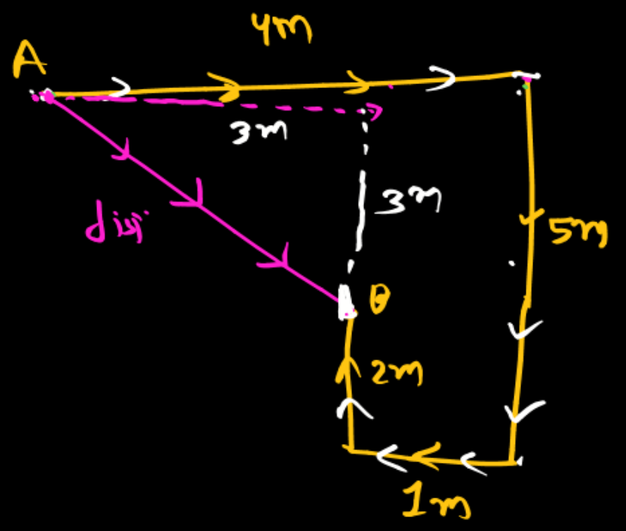
$$\text{disp}^n = (5\hat{i}-2\hat{i}) + (8\hat{j}-4\hat{j}) + 12\hat{k}-7\hat{k}$$

$$\underline{\text{disp}^m} = 3\hat{i}+4\hat{j}+5\hat{k}$$

$$|\text{disp}^m| = \sqrt{9+16+25}$$

$$= \sqrt{50} = \sqrt{25\times 2} = \underline{5\sqrt{2}\ m}$$

(6) dist^n & disp^n b/w 'A' & 'B'



dist^n = 4+5+1+2 = 12m

disp^m = $3\hat{i}-3\hat{j}$

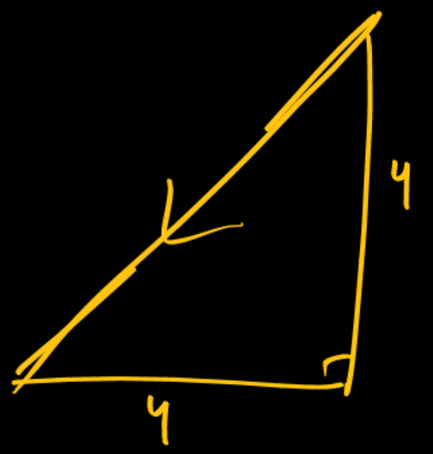
= $\sqrt{9+9}$ = $\sqrt{18}$ = $\sqrt{9\times2}$

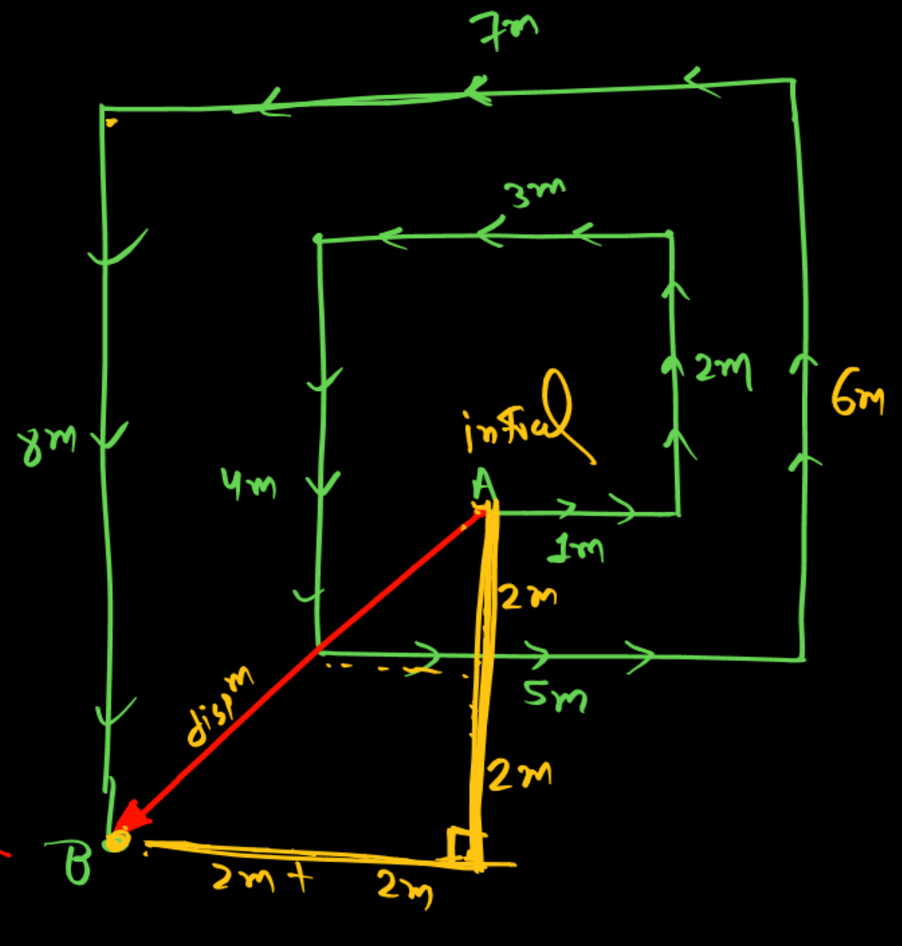
= $3\sqrt{2}$

⑦ MRQ

distance = 36m ✓

disp^m





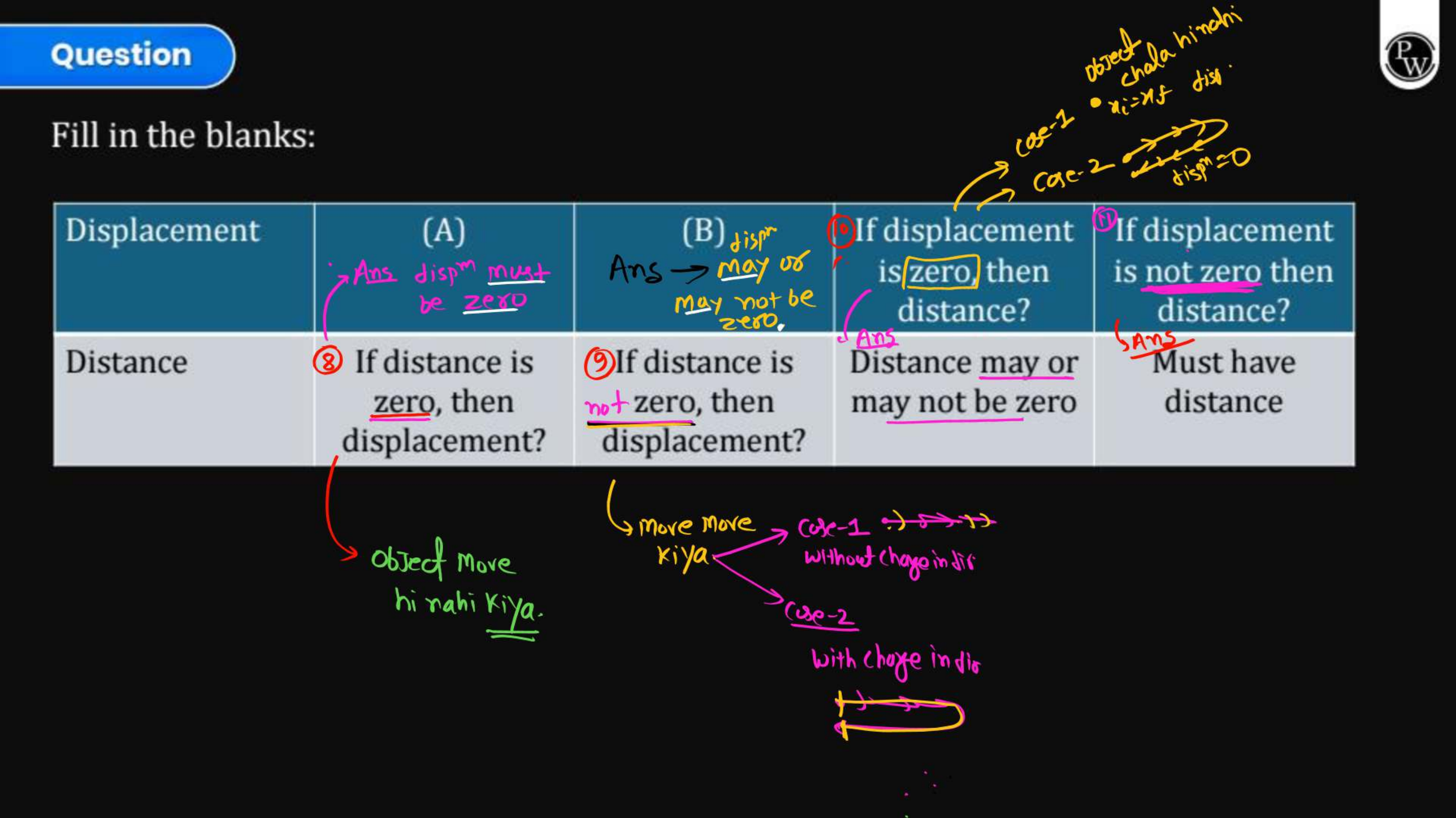
$$\text{disp}^{m} = \sqrt{4^2+4^2}$$

$$= \sqrt{16+16} = \sqrt{32} = 4\sqrt{2}$$

Question

Fill in the blanks:

Displacement	(A)	(B)	(10) If displacement is zero, then distance?	(11) If displacement is not zero then distance?
Distance	(8) If distance is zero, then displacement?	(9) If distance is not zero, then displacement?	Distance may or may not be zero	Must have distance



Question

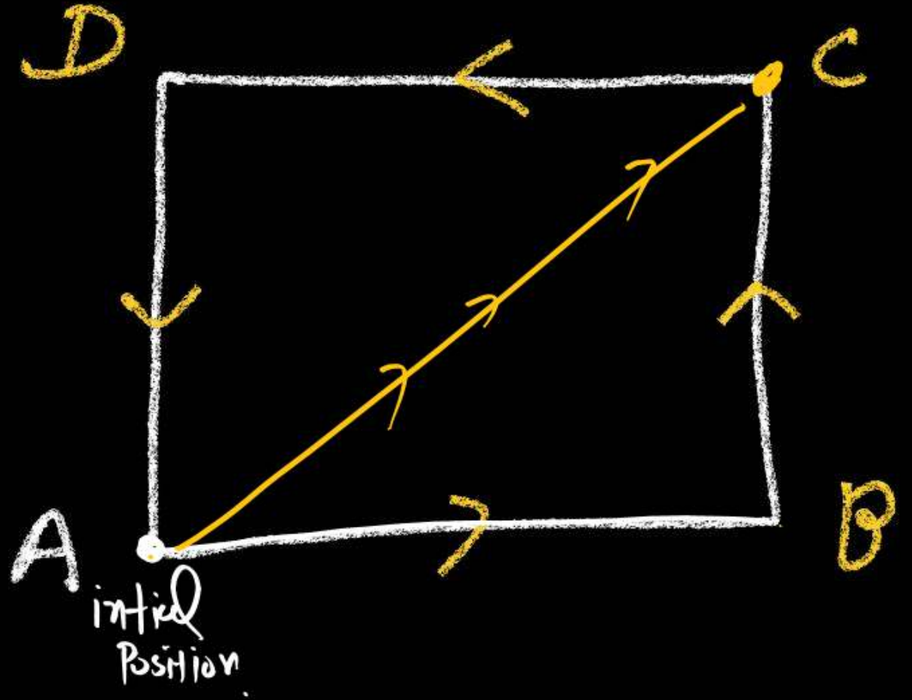
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12

TRUE / FALSE

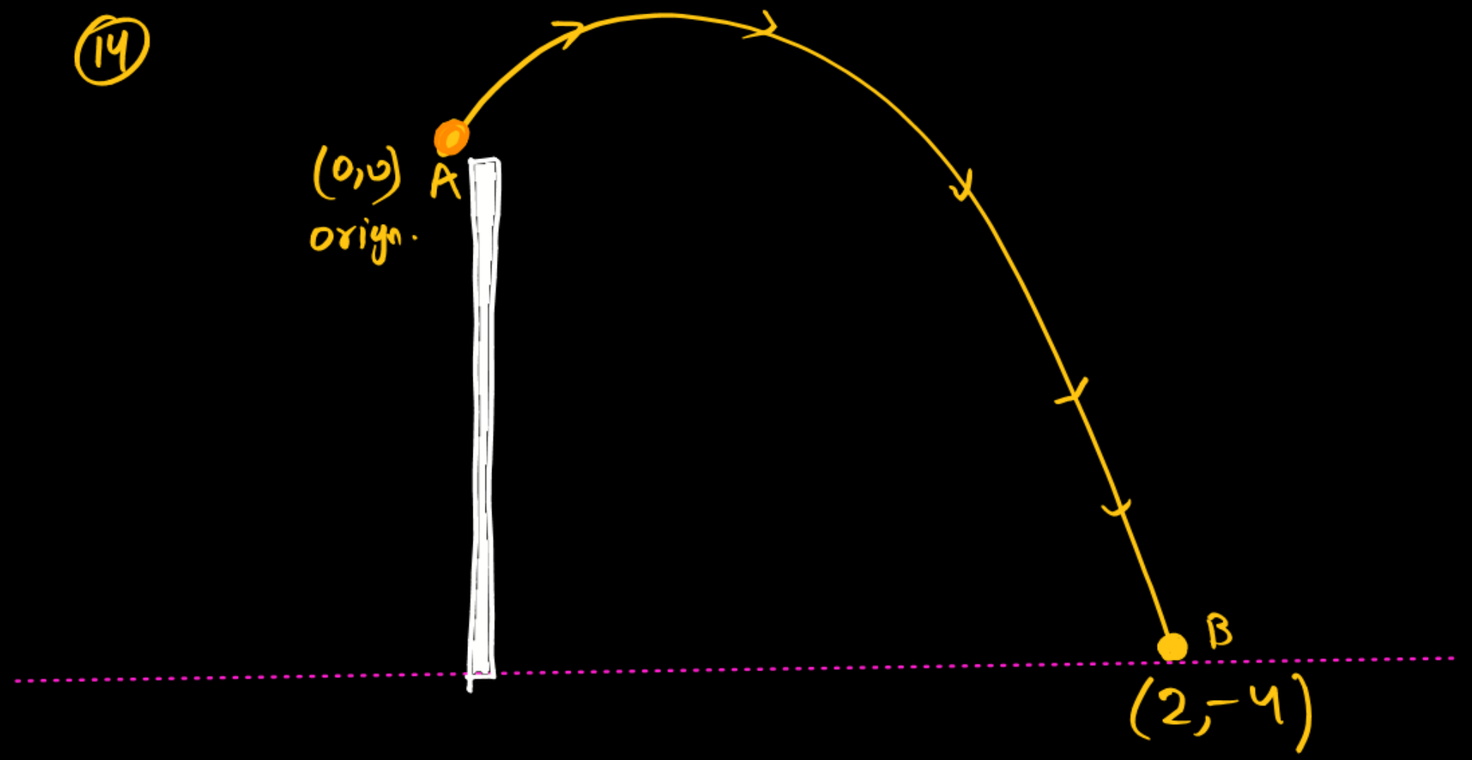
1. Distance must be equal to magnitude displacement in ID motion.
2. If displacement of particle is zero, then distances must be zero.
3. If distance of particle is zero, then displacement must be zero.
4. Magnitude of displacement is always equal to distance.
5. Magnitude of displacement is always equal to distance in ID motion.

(13) find dist^{n} & disp^{m}; object is moving on square park (l)



Path	distance	disp^{n}
A → B		
A → C		
A → D		
A → A	4l	0

(14)



dist^n → can' find

$$\text{disp}^m = \vec{r}_f - \vec{r}_i$$

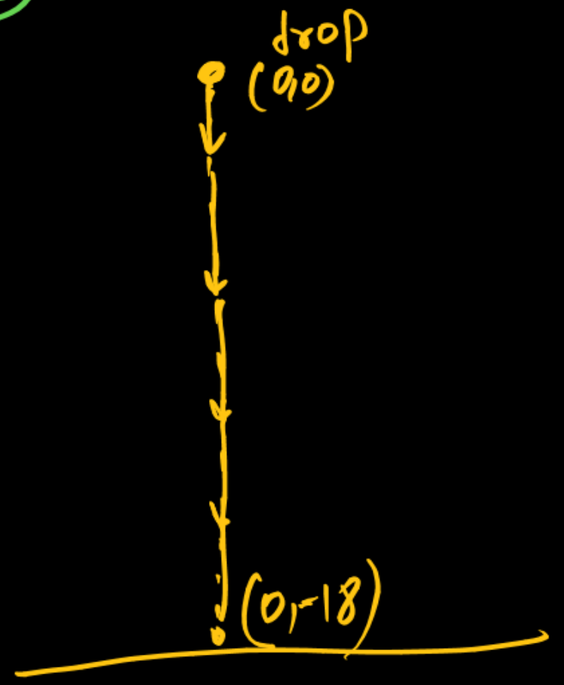
$$= 2\hat{i} - 4\hat{j} - 0\hat{i} - 0\hat{j}$$

$$\vec{s} = 2\hat{i} - 4\hat{j}$$

$$|\vec{s}| = \sqrt{(2)^2 + (-4)^2}$$

$$= \sqrt{20} = 2\sqrt{5}$$

(15)

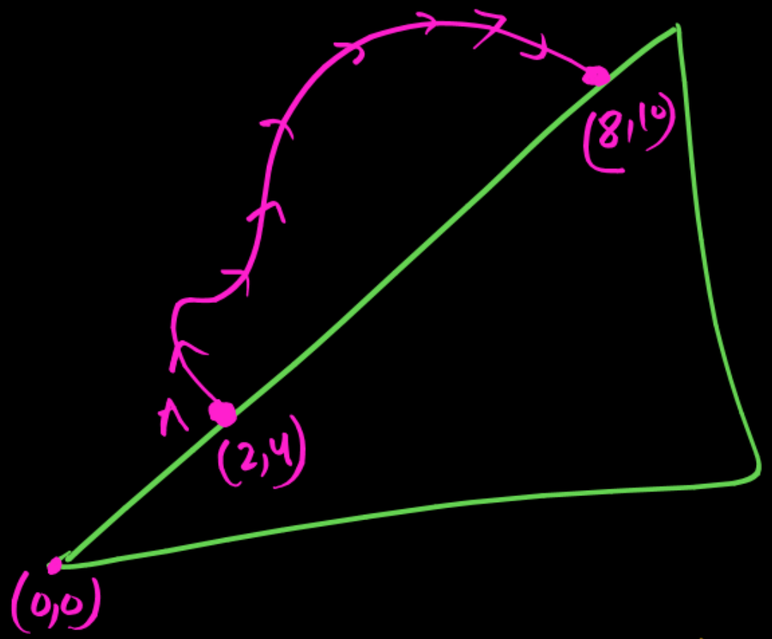


$$\text{disp}^n = -18\hat{j} - 0\hat{i} - 0\hat{j}$$

$$\boxed{\text{disp}^n = -18\hat{j}}$$

$$\text{dist}^n = 18\ m$$

(16)



$$\text{disp}^n = (8\hat{i}+10\hat{j}) - (2\hat{i}+4\hat{j})$$

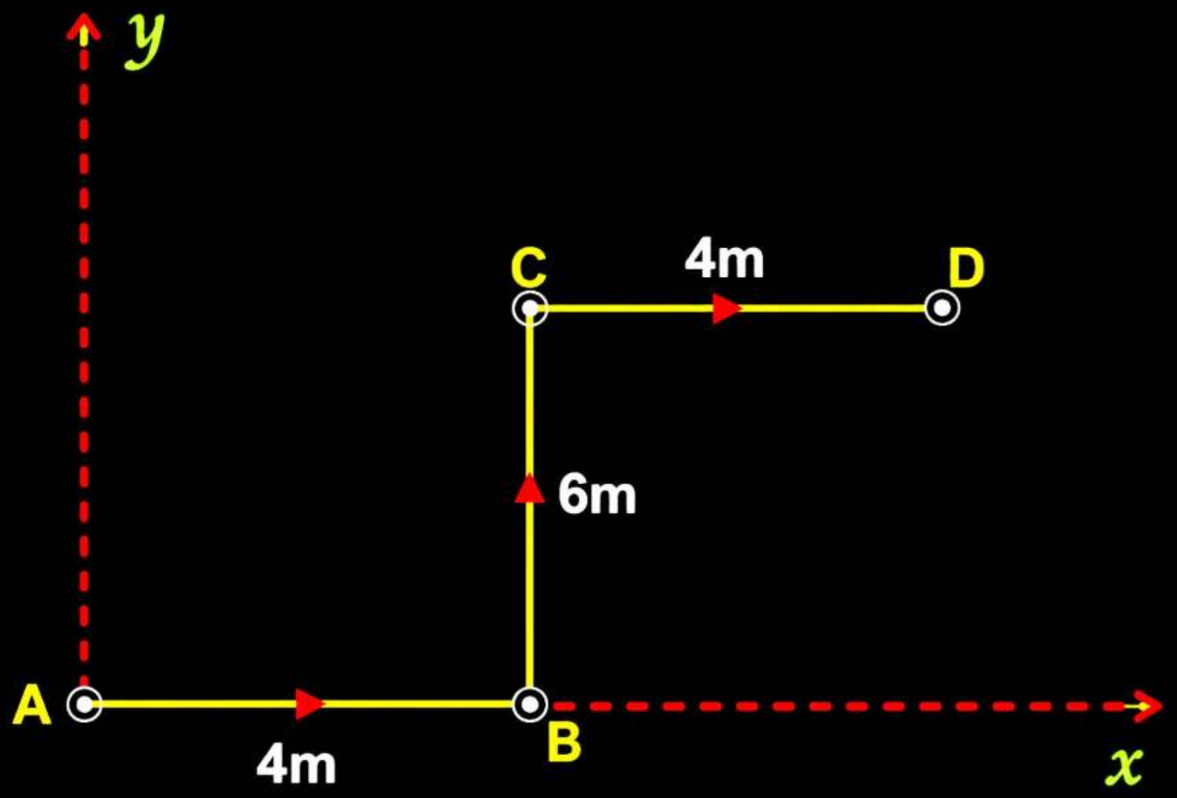
$$= 6\hat{i}+6\hat{J}$$

$$|\text{disp}| = 6\sqrt{2}$$

Question 17

Distance and displacement will be?

1. 14m, 8m
2. 14m, 6m
3. 14m, 10m
4. None



Question

(18) No copy

Correct statement among the following is:

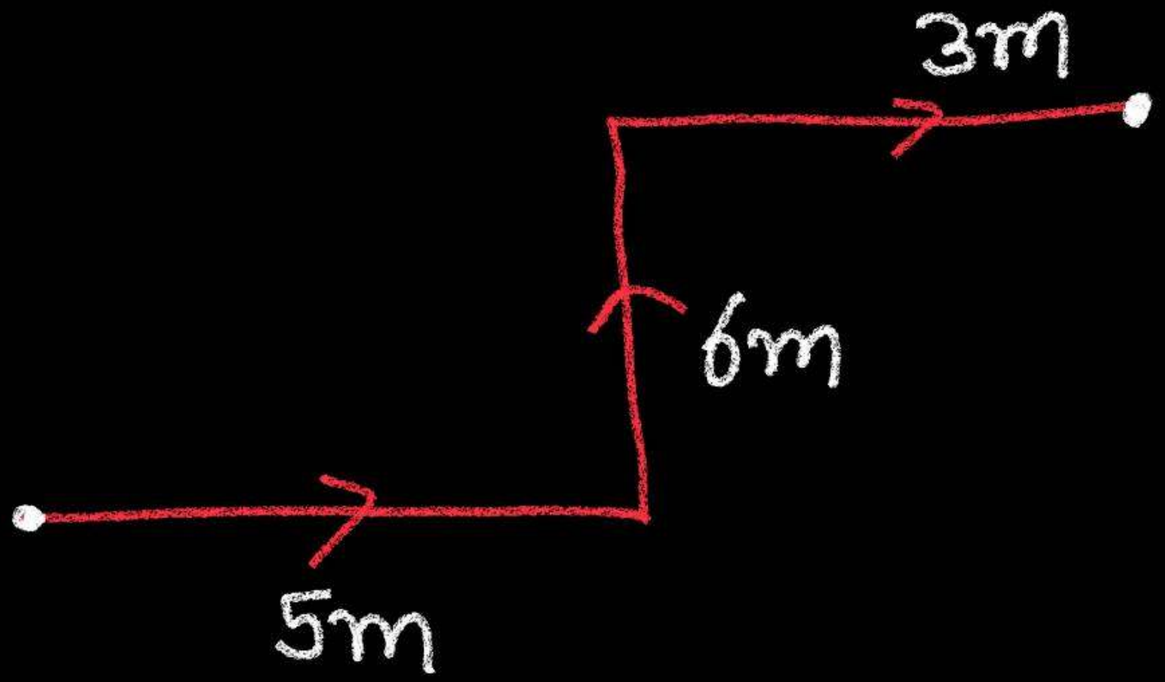
1. When displacement is zero, distance travelled is not zero.
2. When displacement is zero, distance travelled is also zero.
3. When distance is zero, displacement is not zero.
4. Distance travelled and displacement are always equal.

Question ۱۵

If a particle moves from point P(2, 3, 5) to point Q(3, 4, 5). The displacement vector be:

1. $\hat{\imath} + \hat{\jmath} + 10\hat{k}$
2. $\hat{\imath} + \hat{\jmath} + 5\hat{k}$
3. $\hat{\imath} + \hat{\jmath}$
4. $2\hat{\imath} + 4\hat{\jmath} + 6\hat{k}$

(20)



(21) A students moves 10m towards east then 20m South and then $20\sqrt{2}$ North west then find displacement

Rough me solve kr ke

Thank

You